



University of Colombo School of Computing

SCS1311 Internet and Web Technologies

Lab Sheet 01

1. Open a terminal (Windows: Command Prompt; Mac/Linux: Terminal).
 - Run
 - Windows: `ipconfig`
 - Mac/Linux: `ifconfig` or `ip a`
 - Find the IP address of your active network connection.
 - Open a browser and search "what is my IP".
 - Ask a friend for their address from the previous step.
 - Run `ping www.google.com` and look carefully at the output.
 - Run `nslookup ucsc.cmb.ac.lk` and look at the output.
 - i. Compare the address in the terminal with the one the website shows. Are they the same? Search for why.
 - ii. Compare your address with your friend's. What is similar, and what is different?
 - iii. Your address has either four numbers separated by dots or a long mix of letters and numbers. Search for what each style is called.
 - iv. What do ping and nslookup seem to do?
2. Go to <https://web.archive.org>. This is the Wayback Machine, which stores archived snapshots of websites.
 - Type a website address into the box in the middle of the page
 - `bbc.com`
 - `google.com`
 - `ucsc.cmb.ac.lk`, and click *Browse History*.
 - Look at the timeline of years. Choose an early year, or the earliest year available.
 - Find a date with an archived snapshot and open it.
 - Explore the old version of the website. Click a few links.
 - Go back and open a recent snapshot of the same website.
 - Compare the old and recent versions.
 - You can also jump straight to a year by typing an address like this in your browser.
<https://web.archive.org/web/2005/http://www.bbc.com>
 - i. What differences do you see in appearance, content and features?
 - ii. Do all the old links and images still work? Why might some be broken?
 - iii. Search for who invented the World Wide Web and in which year.
 - iv. Where does your oldest snapshot sit on that timeline?
3. Open any simple website. <https://example.com>

- Open the browser's Developer Tools or Inspect option.
 - In Chrome: right-click the page → Inspect
 - In Safari: right-click → Inspect Element (you may need to enable the Develop menu first)
 - In Firefox: right-click → Inspect
 - Look at the different sections and labels you can see.
 - Find the part that contains the text "Example Domain".
 - Find the link on the page.
 - Click the link and observe what happens.
 - Go back to the page and change the size of the browser window. Observe what happens to the page.
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- i. What did you find when you opened Inspect?
 - ii. Could you find the text and the link inside the page?
 - iii. What happened when you clicked the link?
 - iv. What happened when you changed the size of the browser window?
 - v. Search for one unfamiliar word or symbol you saw in the Inspect window and explain what you think it means.